# Sachin Ashok

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LinkedIn  Google Scholar  GitHub

## Research Interests

Broadly interested in networked systems with recent emphasis on enhancing performance, monitoring, and root cause analysis for microservices and cloud systems.

## Education

2021 - **PhD.** in Computer Science; CGPA: 4.0
University of Illinois at Urbana-Champaign
Advisors: **Prof. Radhika Mittal**, **Prof. Philip Brighten Godfrey**

2014 – 2018 **B.Tech.** in Computer Science and Engineering
National Institute of Technology, Trichy

## Publications

**TraceWeaver: Blackbox Request Tracing for Modern Cloud Applications**
**Sachin Ashok**, Vipul Harsh, Brighten Godfrey, Radhika Mittal, Srinivasan Parthasarthy, Larisa Shwartz
*Under review at ACM SIGCOMM, 2024.*

**Murphy: Performance Diagnosis of Distributed Cloud Applications**
Vipul Harsh, Wenxuan Zhou, **Sachin Ashok**, Radhika N. Mysore, Brighten Godfrey, Sujata Banerjee
*In Proceedings of ACM SIGCOMM, 2023.*

**Fast and Efficient Look-Ups via Data-Driven FIB Designs**
**Sachin Ashok***, Aditi Partap*, Ammar Tahir* (* = equal contribution)
*In Proceedings of ACM SIGCOMM FIRA Workshop, 2022.*

**Data-Driven Network Path Simulation with iBox**
**Sachin Ashok**, Shubham Tiwari, Nagarajan Natarajan, Venkata N. Padmanabhan, Sundararajan Sellamanickam
*In Proceedings of ACM SIGMETRICS, 2022.*

**Leveraging Service Meshes as a New Network Layer**
**Sachin Ashok**, P. Brighten Godfrey, Radhika Mittal
*In Proceedings of ACM HotNets, 2021.*

**iBox: Internet in a Box**
**Sachin Ashok**, Sai Surya Duvvuri, Nagarajan Natarajan, Venkata N. Padmanabhan, Sundararajan Sellamanickam, Johannes Gehrke
*In Proceedings of ACM HotNets, 2020.*

**Reinforcement Learning for Bandwidth Estimation and Congestion Control in Real-Time Communications**
Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, **Sachin Ashok**, David Zhao, Ross Cutler, Yan Lu, Johannes Gehrke
*In Proceedings of NeurIPS MLForSystems workshop, 2019.*

## Research Experience

2018 - 2020 **Microsoft Research, India** **Research Fellow**
Mentors: **Venkat Padmanabhan**, **Naga Natarajan**, **Sundar Sellamanickam**, **Johannes Gehrke**

- **iBox: Internet in a Box**: Designed a data-informed network simulator for recreating the behaviour of paths in a target network such as the Internet.
  *Published at SIGMETRICS'22.*

- **R3Net: Reinforcement Learning for Bandwidth Control**: Developed realistic network environments to train RL agents for bandwidth estimation & congestion control.
*Published at MLForSystems'19.*

## Industry Experience

2016 - 2017 **Mozilla Organization** **Intern**
Mentors: **Franziskus Kiefer**, **Tim Taubert**

- **Blake2 support for Mozilla's NSS library**: Implemented a Blake2 (a cryptographic hash function) module for Mozillas Network Security Services (NSS) library. The library is used by Google, RedHat, and Mozilla products like Thunderbird.

2017 **Samsung R&D, India** **Intern**
Mentors: **Pratibha Moogi**, **Karthikeyan Somanathan**

- **Object localization framework**: Developed an end-to-end system to leverage deep learning models using custom convolutional neural networks for localization of objects in a given image.

## Invited Talks

- Distributed Tracing without the Pain!
**KubeCon 2022**, Detroit, USA

- Fast and Efficient Lookups via Data-Driven FIB Designs (presented virtually)
**FIRA@SIGCOMM 2022**, Amsterdam, Netherlands

- Data-Driven Network Path Simulation with iBox
**ACM SIGMETRICS 2022**, Indian Institute of Technology Bombay (IIT-B), India

- Leveraging Service Meshes as a New Network Layer (presented virtually)
**CSL Student Conference 2022**, University of Illinois at Urbana-Champaign

- Leveraging Service Meshes as a New Network Layer (presented virtually)
**HotNets 2021**, University of Cambridge, Virtual Event

- iBox: Internet in a Box (presented virtually)
**HotNets 2020**, University of Chicago, Virtual Event

- iBox: Internet in a Box for Realistic Network Simulation
**Bandwidth Control Workshop 2019**, Microsoft Research, Redmond

## Honors

- Awarded student grant/ invite to attend SIGMETRICS 2022, SIGCOMM 2023, HotNets 2023.
- Invited for student panel discussion at NetworkingChannel.
- Invited for demo at **Microsoft TechFest**, 2020.
- Finalist, **Pragyan CTF**, an international level security contest held by **NIT Trichy**, 2018.
- Among top 100 chosen by the **Indian Institute of Science (IISc)** for **CSA Undergraduate Summer School**, 2017.
- Among top 25 chosen by **Institute of Mathematical Sciences (IMSc)** for **Summer Research Program**, 2017.
- Among 45 chosen worldwide for **Mozilla Winter of Security**, 2016.

## Skills

Languages: `C++, C, Python, Bash`
Tools: `Kubernetes, eBPF, Protobuf, ns-2/3, gRPC`